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The Palaeomagnetism of the Permian Rocks of the Black Forest, Germany

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Summary

Palaeomagnetic measurements made upon Permian samples from the Black Forest, Odenwald and Pfälzerwald (excluding those measurements on samples from the Schopfheim basin) are consistent with those from north-western Europe. The virtual pole based upon 18 sites is 174·9 °E. 48·4 N. The measurements made on the fine-grained shales and mudstones from the Schopfheim basin are significantly different from the preceding. Since incorrect age assignment, sampling, components of secondary magnetic and tectonic effects can be excluded, the difference is attributed to incomplete removal of the effects of secular variation.

The Scharfenstein massif previously regarded as Permian, was found to have a magnetization consistent with a Devonian or early Carboniferous age, and a few dykes in the southern Black Forest had magnetizations consistent with a late Carboniferous or Permian age. The results suggest that both the early and late phase of acid intrusives were sampled.

1. Introduction

Although palaeomagnetic measurements had previously been carried out on Permian rocks from the Black Forest, the samples were few in number and came mostly from the northern part of the massif. The re-investigation of the Permian from the whole of the Black Forest region, supplemented by sampling in the Odenwald and Pfälzerwald, was stimulated by the discovery of small differences in mean directions of magnetization interpreted in terms of tectonic rotations in other areas. There are similarities in the tectonic framework of the Black Forest compared with such regions as the Bohemian Massif or the massif of the Maures-Esterel which prompted the idea of comparing palaeomagnetic results.

As a preliminary to the palaeomagnetic study the geology of the Permian sedimentary basins of the Black Forect, and aspects of the pre- and post-Permian tectonic history will be reviewed. A considerable volume of work concerned with the Black Forest has been published and the review will only refer to those features relevant to the palaeomagnetic study. Reference to some regional hypotheses is included as certain aspects of the tectonic history of the region have been related to events affecting the continent as a whole.

2. Outline of the geological and tectonic history of the Black Forest

The Black Forest of southern Germany is an uplifted part of the pre-Variscan and Variscan basement of Europe. Its western margin is clearly defined by the

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NNE–SSW trending Rhine graben faults which separate it from the Vosges its companion massif to the west. This system of faults was considered by Stille (1923/25) to be part of a major tectonic feature stretching across Europe from the Mediterranean to Lake Mjösen in Sweden (Fig. 3). To the east the Black Forest has no well-defined boundary as the basement disappears under a mantle of Mesozoic sediments (Fig. 1).

The basement of the Black Forest is composed mainly of gneisses in the central area with granites to the north and south (Mehnert 1953, 1958). In the southern part of the Black Forest folded Upper Devonian and Lower Carboniferous sediments occur in the Badenweiler–Lenzkirch zone. The principal folding movements of the Variscan orogeny to affect the Black Forest are attributed by Sittig (1969) to the Bretonic phase, with subsequent Sudetic and probably Asturian movements. One result of this activity was the formation of a number of basins in and around the Black Forest massif (Fig. 2). The basins were subsequently the site of the rapid deposition of clastic material with fanglomerates common but with some intercalations of finer-grained sandstones and shales. In general although these sediments are poorly fossiliferous much of the sedimentation is regarded as Permian in age. In some basins there is evidence that sedimentation began in Stephanian times (Eck 1892;

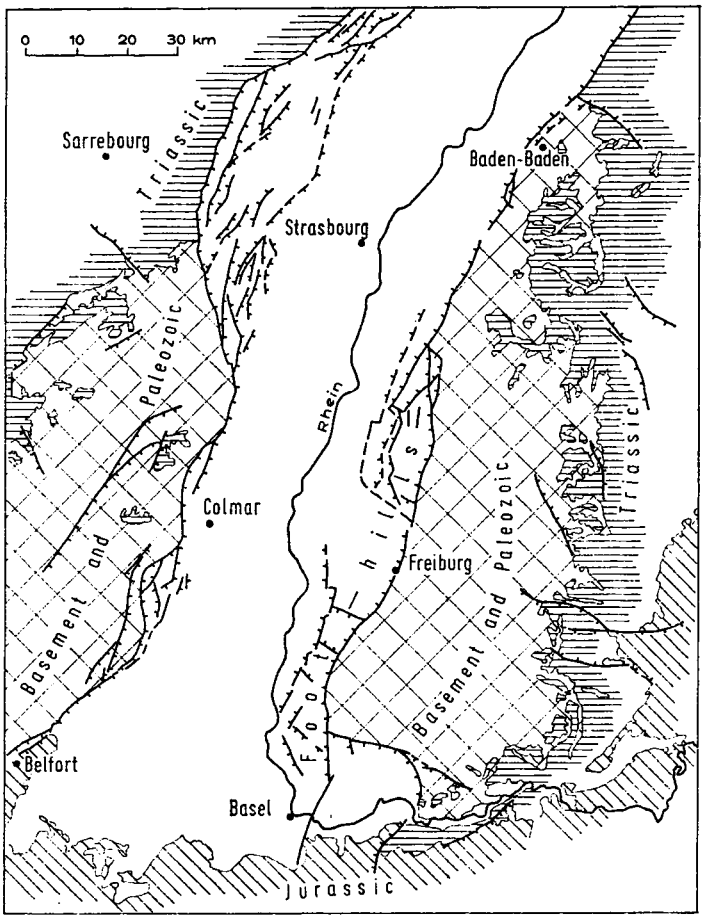


FIG. 1. Sketch map of the Schwarzwald and Vosges massifs delimited by the outcrop of the Buntsandstein and the graben faults. The outcrop of the base of the Jurassic and certain of the principal faults are shown.

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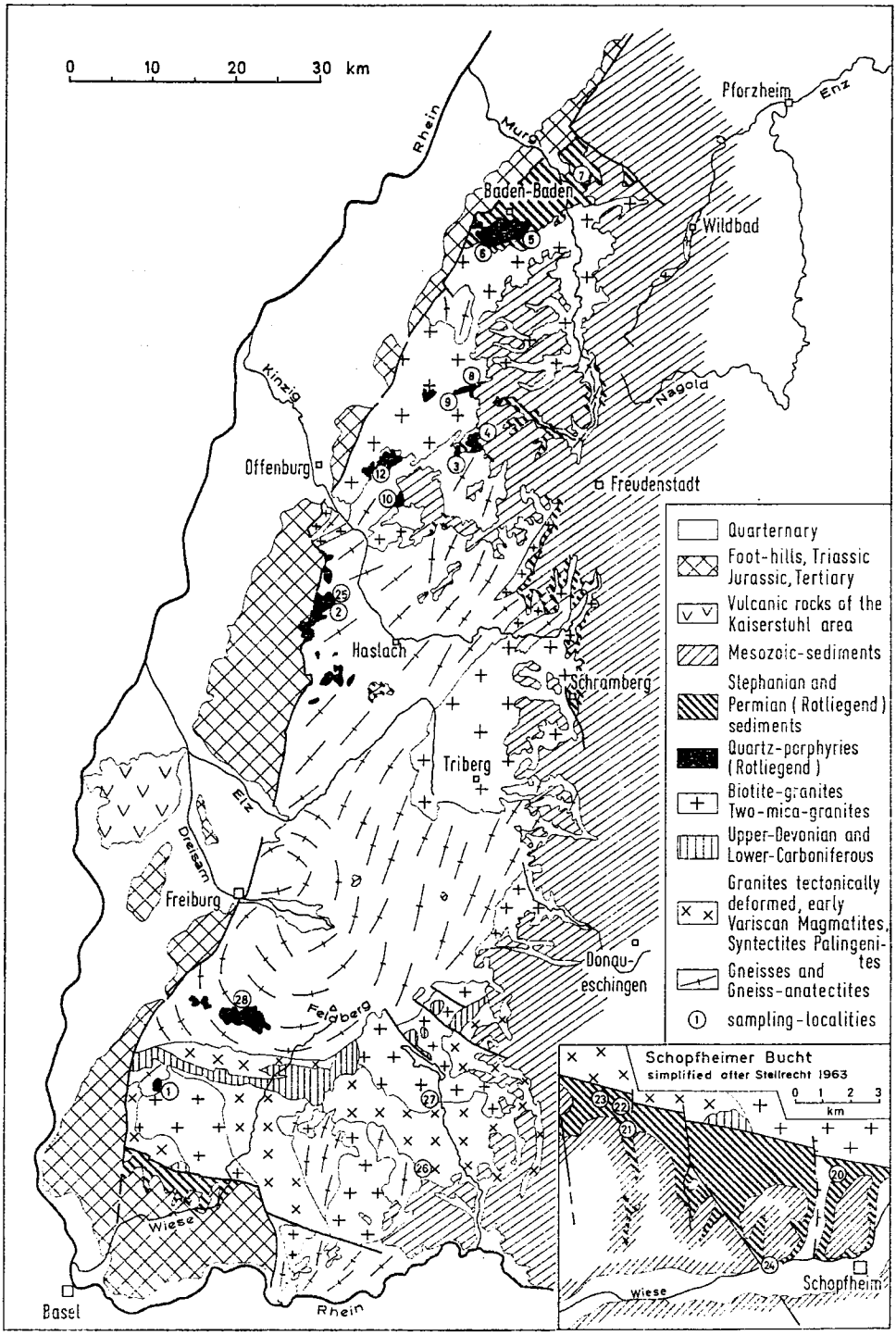


FIG. 2. Geological sketch map of the Schwarzwald (simplified after Metz 1960) showing the Permian outcrops of igneous rocks in black, sediments in oblique ruling. Sampling sites are numbered to conform with numbers in Table 1 and Fig. 4(a) and (b).

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Bräuhäuser 1933). In the Schopfheim basin, of particular concern in this article, it is possible that the earliest sediments (not exposed) may also be of Stephanian age (Stellrecht 1962, 1963a; Schmassmann & Bayramgil 1946, Fig. 2). The whole region was uplifted and eroded during the latter part of the Permian for Lower Triassic sediments were deposited upon a peneplaned surface (Strigel 1922). Further uplift occurred particularly during Plio-Pleistocene times and the renewed erosion has cut through the Mesozoic sedimentary cover exposing the basement rocks and the Permian succession.

There was considerable volcanic activity during Permian times. This is indicated not only by the remains of tuff, flows and ignimbrites (Maus 1967) but also by the large quantitites of volcanic material in some of the conglomerates. The common rock types are porphyries and quartz porphyries. The volcanic activity appears to be restricted to the regions where Carboniferous intrusions occur (Weyl 1938) and in fact the intrusion of the younger granites may have been contemporaneous with the older of the volcanics (see Mehnert 1958). The volcanics are usually associated with unfossiliferous rocks and as no radiometric dating has been carried out their exact age is hard to define. They are generally considered to have occurred throughout the Rotliegendes (Lower Permian). The distribution in space of the Permian outcrops in the Black Forest is shown in Fig. 2.

The final tectonic event in the history of the Black Forest was the formation of the Rhine graben (see Knetsch 1963; Illies 1965, 1967a; Rothé & Sauer 1967) in Cenozoic times. To the south the graben ends against (or is offset in the region of) the Swiss Jura, to the north it continues through Hesse. There have been suggestions of small lateral movements along the fault lines (cf. Knetsch 1963; Wagner 1950, Illies 1965), however the general tectonic trends in the Vosges and Black Forest can be matched with little need for offset (Brinkmann 1923/25; Wagner 1950; Negendank 1969). There have been further suggestions that the location and trend of the graben faults reflects a deeper structural pattern, possibly of Pre-Palaeozoic age (Illies 1962; Knetsch 1963; Wilser 1933), but there is certainly no north–south alignment in the Permian sedimentary basins which clearly reflect their Variscan ancestry, nor do the volcanics in their distribution provide any clear support of the idea (see however Illies 1962). The search to find an earlier history of a Rhenish trend in the Rhine graben is closely related to Stille's (1947) concept of the Mediterranean–Mjösen line as a structural entity as well as a geographical alignment. Richter (1934) went so far as to consider the line as the demarcation between an Ibero-Gallic shield and an Afro-Germanic shield with some relative rotation between the two. However the age of the recorded tectonic activity varies along its length (see Murawski 1960). The Cenozoic tectonic activity in the Rhine valley may still be related to large scale tectonics for Le Pichon (1968) suggests that the abrupt change in the azimuth of faults between 71 °N and 66 °N, implying distortion, may be related to a 'zone of extension traversing Europe, including the Bresse and Limagne in southern France, the Rhine graben and possibly the subsiding zone of the North Sea'. This is the Mediterranean–Mjösen line in a new guise.

3. Palaeomagnetic observations

(*a*) *Sampling*

The attempt was made to obtain samples from all the major Permian outcrops in the Black Forest, both in igneous and sedimentary rocks. The principal difficulties were in sampling the sediments. In general the deposits were predominantly conglomeratic, but in each of the three main basins, the Oos trough in the north near Baden-Baden, the Schramberg trough in the east and the Schopfheim basin in the south, fine-grained horizons occur. The fine-grained lithologies were extremely soft and friable, and no samples were obtained in the single suitable fine-grained rock exposed

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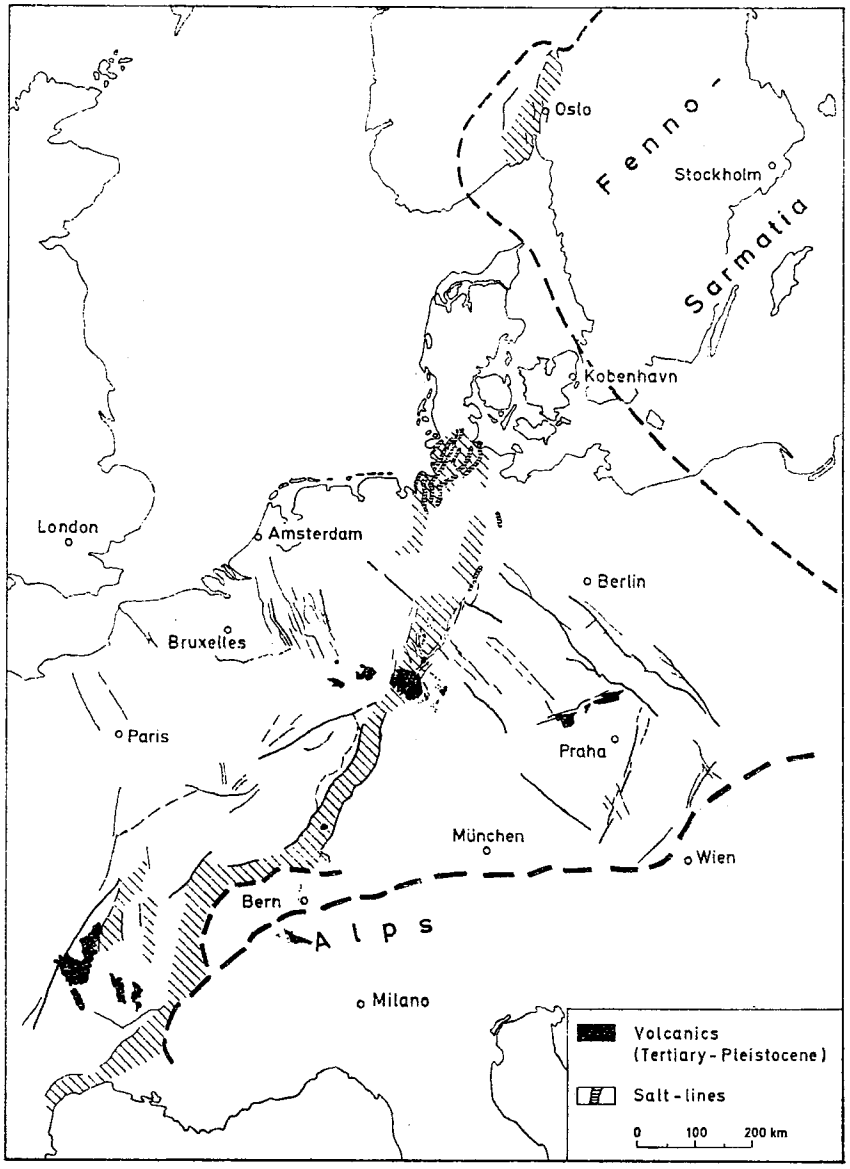


FIG. 3. The principal tectonic lines in Central Europe (Mediterranean–Mjosen zone oblique shading) after Stille (1923–24), Von Zwerger (1948), and the tectonic map of Europe 1 : 2 500 000.

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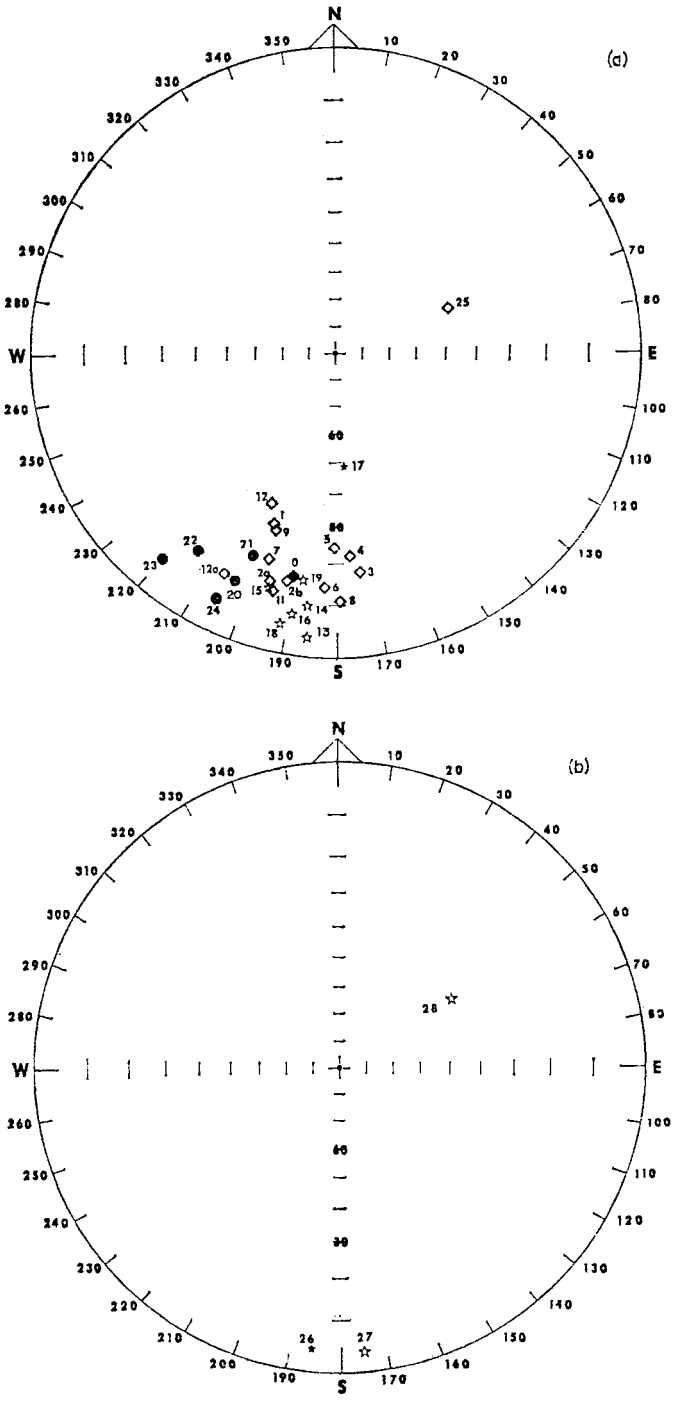


FIG. 4(a). Site mean directions of magnetization. Following convention positive inclinations are shown by a solid symbols, negatives by an open symbols. Wulff projection. ● Schopfheim basin, ◆◇ Black Forest, ☆* Odenwald, Pfalzerwald. (b) Site mean directions of magnetization of the granitic dykes and of the Scharfenstein quartz porphyre.

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in Schramberg trough and very few in the Oos trough. The most extensively sampled horizon in the Schopfheim basin was the upper fine grained red shale r_{03} whose outcrop has been mapped by Stellrecht (1963b). The distribution of sites is shown in Fig. 2 upon a map simplified from the publication. The maximum thickness of r_{03} is about 100 m althought at any one outcrop less than 10 m is ever exposed. Closely spaced sampling at two outcrops was possible where 1–2 m were exposed, at the other outcrops as many samples as possible were collected, one per bed.

The igneous rocks were in general better exposed and numerous quarries exist where the rock is, or was, worked for use as roadmetal or building stone. All the principal types were sampled. They showed a considerable variety in colour, from near white (presumably altered during the late hydrothermal phase) to reddish and purple. The texture of the rocks shows generally a glassy groundmass. In some outcrops well developed fluidal texture textures showed quite complicated flow patterns. Both intrusions and flows occur, and from each unit a minimum of six samples was collected. In only one or two cases did there appear to be more than one unit exposed in any given outcrop, and at only one quarry was it possible to sample two different units. The distribution of the sites is indicated in Fig. 2, the site numbers corresponding to the entries in Table 1.

(*b*) *Sample preparation and measurement*

Single cylinders were drilled from each of the samples collected. A number of the more friable sediments disintegrated during this process. Usually a single cylinder long enough to provide two cores was obtained. Few of the cores had high intensities of magnetization and all were measured using a Foster fluxgate spinner. The intensity of NRM of the igneous rocks was in the range $0{\cdot}1-30\times10^{-5}$ emu cm^{-3}, that of the

Table 1

Summary of measurements on rocks from the Black Forest and adjoining regions. Under the heading sampling; (a) represents the number of samples used in the computation of the statistics, and (b) represents the number of samples collected.

	D	*I*	Sampling (a)/(b)	α_{95}	*K*	*R*	Demag.
Black Forest: Schopfheim Basin sediments							
20 Enkenstein	203·1	+11·2	7/8	11·4	29	6·792	500 °C
Wieslet	—	—	—	—	—	—	—
Schlächtenhaus	—	—	—	—	—	—	—
21 Unterer Schlangen-grabenweg	201·6	+18·5	16/17	13·1	9	14·309	550 °C
22 Waldparkplatz Scheideck	214·3	+13·5	1/3	—	—	—	500 °C
23 Roter Rain	220·3	+7·0	6/6	15·9	19	5·733	500 °C
24 Maulburg–Scheinberg	205·3	+6·5	9/11	15·5	12	8·336	500 °C
Mean of site means	209·0	+11·4	5 sites	8·8	76	4·947	
Mean of samples	205·9	+12·4	38 samples	7·1	12	34·798	
Virtual pole position	154·1 °E	30·7 °N	dp 4·6	dm 8·9	(sites)		
Black Forest: North of Schopfheim Basin, sediments, extrusives and intrusives							
1 Stockberg SW of Badenweiler	200·0	−28·5	11/16	9·6	24	10·580	150 Oe
8 Bosenstein near Seebach qu. porphyre	179·7	−11·3	7/7	2·4	642	6·991	150 Oe
9 Edelfrauengrab, same porphyre as above	198·2	−26·6	7/7	13·6	21	6·709	150 Oe
10 Gengenbach, Forst-köpfle qu. porphyre	190·3	+16·3	6/6	14·2	23	5·786	NR

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Table 1—*Continued*

	D	*I*	Sampling (a)/(b)	α_{95}	*K*	*R*	Demag.
10a Same porphyre 700 m west	no result		—	—	—	—	—
11 Between Schweighausen and Welschau	194·5	−12·6	7/7	10·2	36	6·835	77 Oe
12 Durbach flow near Geisberg	203·0	−33·7	5/7	10·5	54	4·926	150 Oe
12a Same flow 700 m NW am Neuweg	206·2	−11·1	3/7	53†	—	—	—
2a Schönberg/Lahr intrusion	195·9	−13·5	6/8	30·0	6	5·152	150 Oe
2b Schönberg/Lahr ignimbrite	191·5	−14·9	6/6	11·2	37	5·870	150 Oe
25 Geroldseck? ignimbrite	67·2	−47·0	5/7	18·9	17	4·769	77 Oe
— Rossbronnen/Reichenbach	no result		—	—	—	—	—
3 Hauskopf/Oppenau ? ignimbrite	173·2	−18·7	6/7	4·9	186	5·970	77 Oe
4 Rotenkopf flow banded ignimbrite	175·6	−22·7	6/6	9·5	51	5·900	150 Oe
7 Michelbach Ro sediments	198·0	−19·0	2/3	estimated mean			
5 Geroldsau Baden-Baden qu. porphyre	180·7	−24·9	7/7	4·0	228	6·974	150 Oe
6 Yburg, Baden–Baden	183·0	−14·8	5/7	15·2	26	4·848	150 Oe
Mean of site means	189·2	−17·9	12	8·7	26	11·758	
Virtual pole position	173·9 °E	49·9 °N	dp	4·7	dm	9·0	
Black forest: various St. Blasien porphyritic microgranite	no result						
26 Föhrenbachmühle/ Nöggenschwiel, quartz porphyre	186·1	+4·3	7/7	14·2	19	6·700	150 Oe
27 Giersbacherkopf, two quartz porphyre dykes	174·6	−3·9	6/6	11·7	12	5·570	300 Oe
28 Scharfenstein, mean of 5 sites qu. porphyre	58·0	−43·2	—	5·4	200	4·980	150 Oe
	(virtual pole 137·9 °E 0·0 °N do 4·2 dm 6·7)						
Pfälzer Wald							
13 Waldhambach, $R_{o/u}$ melaphyre	185·9	−3·4	8/8	0·6	8	7·140	150 Oe
14 Klingenbachtal/ Klingenmünster, melaphyre	186·8	−10·3	5/6	9·7	63	4·937	150 Oe
15 Albersweiler, $R_{o/u}$ melaphyre	195·3	−13·3	5/6	18·8	17	4·771	150 Oe
16 Annweiler sediment	189·6	−8·0	3/3	10·7	134	2·985	550 °C
17 Sarnstall bei Annweiler	175·3	+49·2	3/5	34·3	14	2·857	550 °C
Mean of site means (excl. 17)	189·4	−8·8	4	6·7	187	3·984	
Virtual pole	174·7 °E	45·2 °N		dp 3·4	dm 6·7		
Odenwald							
18 Dossenheim	191·4	−5·3	6/7	16·6	17	5·711	300 Oe
19 Schriesheim	188·0	−16·0	2/4	estimate			
Overall mean direction of sample means from the Black Forest, Pfälzer Wald Odenwald (excluding Schopfheim Basin)	189·3	−15·0	16·0	6·0	35	17·508	
Virtual pole position	174·1 °E	48·4 °N	dp 3·1	dm 6·1			
Schopfheim mean	209·0	+11·4	8·8				
virtual pole	154·1 °E	30·7 °N	dp 4·6	dm 8·9			

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sediments $1{\cdot}0-100\times10^{-5}$ emu cm^{-3} although the stable magnetic component was often only one tenth or less of the NRM value (see Figs 5 and 6).

At least one core from each sample was magnetically cleaned. All the volcanic rocks were subjected to alternating field demagnetization, the sediments and a few duplicate cores from a number of volcanic sites underwent thermal demagnetization. All the demagnetization experiments were carried out in a mu-metal room . A pair of ancillary coils around the furnace reduced the residual field to less than 40γ during thermal demagnetization. Thermal demagnetization was carried up to 500° C in 100° C steps. Alternating field demagnetization samples were only treated at two or three steps, usually at 77, 150 or 300 Oe. The power source for the alternating field was a 400 c/s generator. During demagnetization samples were rotated about three axes. Illustrations of direction changes are given in Fig. 5, with normalized demagnetization curves in Fig. 6.

Calculation of the site mean directions of magnetization and the Fisher statistics were carried out on a Wang computer. Although it is possible to tell when secondary components of magnetization have been removed by diagrams such as those illustrated

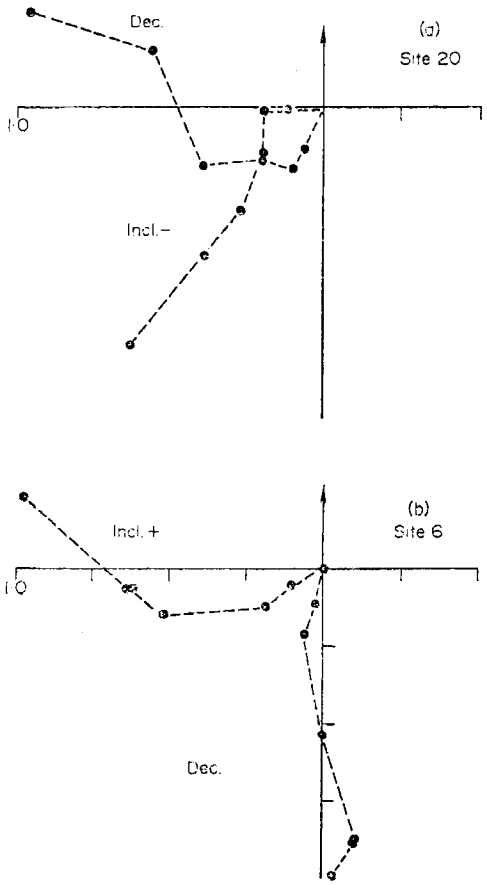


FIG. 5. Changes in declination and inclination plotted against normalized intensity. Site numbers correspond to numbers in Table 1 and curves in Fig. 6. (a) Thermal demagnetization of a sediment, points every 100° to 500° C final stage of 550° C. (b) A.C. demagnetization of a lava stages NRM, 77, 150, 300, 600 1200 Oe.

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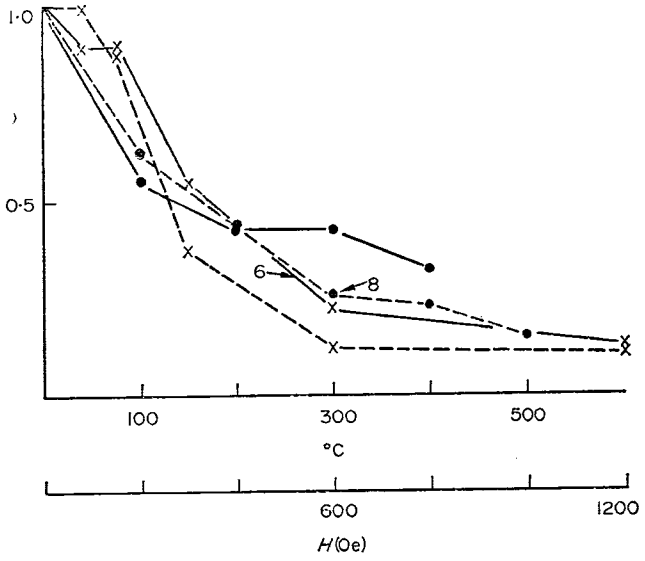


FIG. 6. Normalized A.C. and thermal demagnetization curves. Site numbers correspond to Table 1. Solid lines samples from site 6, broken lines samples from site 8. W Thermally demagnetized specimens (use upper scale); + A.C. demagnetized specimens (use lower scale).

in Fig. 5, the direction regarded as best representating the orginal field direction at the time of formation of the rock was selected from the minimum circle of confidence-maximum value of K, given by the Fisher statistics. These results are illustrated in Fig. 4(a) and (b) and listed with appropriate parameters in Table 1. The site mean direction of magnetization for the sediments was obtained by giving each site unit weight despite the differences in the numbers of samples per site. If the sedimentation rate at the different sites can be regarded as essentially the same the averaging technique avoids giving too much weight to the two sites where there was dense sampling of a restricted interval. The mean giving each sample unit weight is also given.

4. Discussion of results

(*a*) *The Schopfheim and northern Black Forest results*

The results from the Black Forest (excluding the Schopfheim basin), Odenwald and Pfälzerwald are consistent (see Fig. 4(a), and Table 1). The virtual pole position from the combined site mean result lies somewhat east of other results from the Permian of northern Europe (Table 2). The data presented in Table 2 is from a number of widely-dispersed regions, it is not intended to be comprehensive. The confidence limits of the Black Forest virtual pole position overlaps some of the results given. The results from the Black Forest (excluding the Schopfheim basin), Odenwald and Pfälzerwald are therefore regarded as typical of northern Europe at this period of time, and imply that the palaeogeographic relationship of the areas concerned has remained constant since that time.

The mean direction of magnetization of the sediments in the Schopfheim basin is distinctly different from the mean direction of magnetization of those rocks lying to the north. The difference cannot be attributed to secondary components of magnetization, for all samples were magnetically cleaned (Fig. 5(a)). It is not an artifact of collecting for although observations could not be made directly with respect to sedimentary bedding planes but required the application of a local dip correction,

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Table 2

Summary of certain Permian and Stephanian data from various locations in northern Europe

	D	*I*	aJJ	Virtual	Pole	dp	dm
Black Forest, Pfälzerwald, Odenwald	189·3	−15·0	6·0	174·1 °E	48°4 °N	3·1	6·1
Schopfheim Basin	209·0	+11·4	8·8	154·1 °E	30·7 °N	4·6	8·9
Krakow volcanics (Birkenmajer & Nairn 1964)	205	−20·0	8·1	164 °E	45°N	5·0	9·0
Vrchlabi (Bucha *et al.* 1963)	205·2	−9·2	4·7	162 °E	45 °N	5·0	9·0
Oslo (1/77)†	202	−33·0	9·5	161 °E	45 °N	6·0	11·0
Exeter Traps (9/89)	198	−25·0	—	149 °E	50 °N	4·0	8·0
(9/90)	188	−13·0	—	165 °E	46 °N	4·0	7·0
Ayrshire (1/75)	187	−6·0	12·0	167 °E	37 °N	6·0	12·0
Inner Sudentic (10/111) Stephanian	195·1	−1·8	3·9	176·8 °E	38·7 °N	2·0	4·0
Plzen Basin)10/112) Stephanian	214·7	+2·1	5·1	151·7 °E	31·1 °N	2·6	5·1

† Numbers in parentheses refer to the listing by Irving or McElhinny in *Geophysical Journal.*

the latter was observed in the field. The correction required to remove the difference found with respect to rocks in the northern Black Forest is geologically exceedingly improbable.

The possibility that the Schopfheim sediments were not Permian in age was considered. The basins around the Schwarzwald which received late Palaeozoic sediments were formed by the Variscan movements (see Section 2) and in the north, Stephanian sediments were laid down. In the southern Black Forest however the dating of the Schopfheim sediments rests upon the recognition of an *Amblypterus* specimen from the Wintersingen bore (see Stellrecht 1963b), and geological opinion appears to be firm in regarding the assignment of an r_{03} age to the sediments sampled as reliable.

Tectonic rotation as an interpretation seems improbable. It is clear from the review in Section 2, that despite a variety of theories and suggestions, there is no evidence that the last phase of tectonic activity to affect the Black Forest, the formation of the Rhine graben, was associated with any significant rotational movement. All the available evidence points to normal faulting. As the Black Forest basement rocks can be traced southwards under the Swiss Jura it is hard to consider the relative rotation of a small segment such as the Schopfheim basin along the faults which bound it. Although such a relative rotation of 10° along the faults cannot be directly excluded, rotation fails to erase the difference in inclination.

An alternative interpretation may be that despite the thickness of beds sampled (up to 100 m of fine grained sediments) the time represented is relatively short, that is, secular variation has not been averaged out. The implication, to the authors, is that the time involved was too short to average out longer term non-dipole effects. Such effects vanish when sampling covers a wider time range as in the regions to the north.

Roche (1958) drew attention to the fact that upon averaging, certain of his Miocene results did not provide a virtual pole coincident with the present geographic poles, and more recently Wilson (1970) has considered certain permanent aspect of the Earth's non-dipole field in upper Tertiary times. While the interpretation is tentative, the authors consider that the best interpretation of the Schopfheim basin sediments is to regard them as results from which the effects of the longer term non-dipole components have not been removed in the averaging process since sampling and treatment can be excluded as a cause, and tectonic rotation seems improbable.

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(*b*) *The Scharfenstein quartz porphyre*

The Scharfenstein quartz porphyry is shown on the Geologisch Petrographische Übersichtskart des Südschwarzwaldes (Metz & Rein 1957) as being of Permian age although distinguished petrologically from the Stockberg type also of Permian age. It is described as a biotite rich ' Deckenporphyr ' and has a strongly altered groundmass and some secondary silicification as a result of subsequent hydrothermal activity. It is not associated with any Permian sediments and the age assigned rests heavily upon petrological similarities (Weyl 1938). More recently Maus (1967) interprets the Scharfenstein quartz porphyry as an ignimbrite with sedimentary inclusions. The inclusions are similar to local sediments dated as latest Devonian or early Carboniferous (Illies, private communication).

When first sampled for palaeomagnetic work the results obtained although consistent could not be interpreted as belonging to rocks of Permian age. Subsequently a total of five sites were collected in different parts of the massif. The results confirmed those from the first samples, and by inspection of the virtual pole position in comparison with results from Western Europe generally, it seems probable that a late Devonian or early Carboniferous age can be assigned to the Scharfenstein quartz phorphyre.

(*c*) *Acid intrusives of St Blasien and district*

The preliminary results from the Scharfenstein massif suggested that a comparison with certain other of the acid intrusives found in the general vicinity of St Blasien and the Schluchsee might be of interest, for Mehnert (1958) has shown the Variscan granites fall into two age groups, those dating from 280 to 310 My and those in the age range 250 to 270 My.

Unfortunately no results were obtained from the granites in St Blasien, however a total of three quartz porphyre dykes near Giersbacherkopf and Föhrenbachmühle provided directions of magnetization (Table 1) consistent with the Permian from the rest of the Schwarzwald (excluding the Schopfheimer Bucht). The number of units sampled is insufficient to draw firm conclusions, but would suggest that these dykes belong to the younger Variscan phase of plutonism (and may be Stephanian or Permian in age) as indicated by Metz & Rein (1958).

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References

Birkenmajer, K. & Nairn, A. E. M., 1964. Palaeomagnetic Studies of Polish Rocks 1. The Permian Igneous Rocks of the Krakow District and some results from the Holy Cross Mountains, *Rocn. pol. Tow. geol.*, **36**, 225–244.

Birkenmajer, K., Krs, M. & Nairn, A. E. M., 1968. A palaeomagnetic study of Upper Carboniferous Rocks from the Inner Sudetic Basin and the Bohemian Massif, *Bull. geol. Soc. Am.*, **78**, 589–608.

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Bräuhäuser, M., 1933. *Erläuterungen zur geologischen Spezialkarte von Württemberg*, Blatt Schramberg (Nr. 129) (W. Kohlhammer) Stuttgart.

Brinkmann, R., 1923/25. Die rheinische Richtung im vorkimmerschen Süddeutschland *Abh. preuss. geol., Landesanst, NF*, **95,** 133–148.

Brooks, C., Wendt, I. & Harra, W., 1968. A two-error regression treatment and its application to Rb-Sr and initial Sr 87/Sr 86 ratios of younger Variscan granitic rocks from the Schwarzwald massif, south-west Germany, *J. geophys. Res.*, **63,** 6071–6084.

Bucha, V., Chlupacova, M., Krs, M., Nairn, A. E. M. & Větrovská, B., 1963. Palaeomagnetic investigations of rocks in Czechoslovakia from Devonian to Quaternary, *Geofys. Sb. C.S.A.V.*, No. **187,** 293–34 (in Russian).

Cornwell, J. D., 1967. The palaeomagnetism of the Exeter Lavas, *Geophys. J. R. astr. Scr.*, **12,** 181–196.

Du Bois, P. M., 1957. Comparison of Palaeomagnetic results for selected rocks of Great Britain and North America, *Phil. Mag. Suppl., Adv. Phys.*, **6,** 177–186.

Eck, H., 1892. Geognostische Beschreibung der Gegen von Baden-Baden, Rotenfels, Gernsbach und, Herrenalb, *Abh. K. preuss. geol. Landesanst*, N. F. 6.

Hoenes, D., 1937. Gesteine und Erzlagerstätten im Schwarzwälder Grundgebirge zwischen Schaunisland, Untermunstertal und Belchen, N. *Jb. Miner., B,* **72 A,** 263–346.

Illies, H., 1962. Oberrheinisches Grundgebirge und Rheingraben, *Geol. Rdsch.*, **52,** 317–332.

Illies, H., 1965. Bauplan and Baugeschichte des Oberrheingrabens, Oberrhein. *Geol. Abh.*, **14,** 1–54.

Illies, H., 1967a. Ein Grabenbruch im Herzen Europas, Geogr. Rdsch., **19,** 281–293.

Illies, H., 1967b. Gestalt und Fundament des Oberrheingrabens im Modell ein Beitrag zur Taphrogenese, *Oberrhein, geol. Abh.*, **16,** 1–10.

Knetsch, G., 1963. *Geologie von Deutschland*, pp. 386 Enke Verlag, Stuttgart.

Le Pichon, X., 1968, Seafloor spreading and continental drift, *J. geophys. Res.*, **73,** 3661–3697.

Maus, H-J., 1967. Ignimbrite des Schwarzwaldes, N. *Jb. geol. Paläont. Mh.*, 461–489.

Mehnert, K. R., 1958. Die geologische Entwicklung des Schwarzwälder Grundgebirges unter Berucksichtigung neuer absoluter Altersbestimmungen, *Z. d. geol. Ges.*, **110,** 2–3.

Mehnert, K. F., 1953. Petrographie und Abfolge der Granitisation im Schwarzwald, N. *Jb. Miner. Abh.*, **85.**

Metz, R. & Rein, G., 1958. Erlauterungen zur Geologisch-petrographischen Übersichtskarte des Südschwarzwaldes.

Murawski, H., 1960. Das Zeitproblem bei der Tektogenese eines Grossgrabensystems, *Nobtizbl. Hess. Landesamt. Bodenforsch.*, **88,** 294–241.

Nairn, A. E M., 1960. Paleomagnetic results from Europe, *J. Geol.*, **68,** 285–306.

Negendank, J. F. W., 1969. Über permische und tertiäre Magmatite im Untergrund des Mainzer Beckens, *Geol. Rdsch.*, **58,** 502–512.

Richter, G., 1934. Das Rheinische Element im Bilde Westeuropas, *Nachr. Ges. Wiss. Göttingen, math.-phys.* Kl., IV, N.F., **1,** 23–38.

Roche, A., 1958. Sur les variations de directions du champ magnetique terrestre au cours du Quaternaire, *C.r. Acad. Sci., Paris,* **246,** 3364–3368.

Rothe', J. P. & Sauer, K., 1967. The Rhinegraben progress report, 1967, *Abh. geol. Landesant.* Baden-Würt., **6,** 1–148.

Rutten, M. G., Everdingen, R. O. & Zijderveld, J. D. A., 1957. Palaeomagnetism in the Permian of the Oslo graben (Norway) and of the Esterel (France), *Geol. e Mijn.*, **19,** 193–195.

Downloaded from https://academic.oup.com/gji/article/27/4/369/772350 by Curtin University Library user on 06 May 2026

Schmassmann, H. & Bayramgil, O., 1946. Stratigraphie, Petrographie und Paläogeographie der Perm-Formation im Schweizerischen Tafeljura und die Steinkohlenfrage der Nordschweiz, *Tätigkeitsber. naturforsch. Ges. Baselld.*, **15,** 9–117, **15**, *Abb., Liestal.*

Sittig, E., 1969. Zur geologischen Charakterisierung des Moldanubikums am Oberrhein (Schwarzwald) *Oberrhein. Geol. Abh.*, **18,** 9–6.

Stellrecht, R., 1962. Stratigraphische und tektonische Fragen am Südende des Schwarzwaldes, *Z. dt. geol. Ges.*, **113,** 607.

Stellrecht, R., 1963a. Zur Stratigraphie und Tektonik der Weitenauer Vorberge am Südostende des Oberrheingrabens, *Oberrhein, geol. Abh.*, **12,** 119–127.

Stellrecht, R., 1963b. Über das Oberrotliegende im Gebiet der Schopfheimer Bucht, *Jber Mitt. oberrhein. geol. Ver.*, **45,** 1–11.

Stille, H., 1923–25. Rheinische Geibirgsbildung im Kristianagebiet und in Westdeutschland, *Abh. preuss. geol. Landesant.*, N.F., **95,** 110–132.

Stille, H., 1923–25. Die saxonischen Brüche, *Abh. preuss. geol. Landesanst*, N.F., **95,** 149–207. Berlin.

Stille, H., 1947. Betrachtung zum Werden des Europäischen Kontinents, *Z. d. geol. Ges.*, **97,** 7–29.

Strigel, A., 1922. *Zur Palaeogeographie des Schwarzwaldes. Die Abrasionsfläche als klimatisch-tektonisches Problem des oberen Perms*, pp. 155, Ehrig, Frankfurt–Heidelberg.

Wagner, G., 1950. Zum Grassbau der Oberrheinlande, Decheniana, 104, 1–10.

Weyl, R., 1938. Die Entwicklung des rotliegenden Vulkanismus im Schwarzwald, *Z. d. geol. Ges.*, **90,** 367–380.

Wilser, J. L., 1933. Palaeogeographie der Perm-Trias-Profile am südöstlichen Rand des Rheintalgrabens, *Naut. Jb. Miner., Abh.*, **71** B, 1–13.

Wilson, R. L., 1970. Permanent aspects of the Earth's non-dipole magnetic field over Upper Tertiary times, *Geophys. J. R., astr. Soc.*, **19,** 417–437.

Zijderveld, J. D. A., 1967. The natural remanent magnetizations of the Exeter volcanic traps (Permiana, Europe), *Tectonophysics*, **4,** 121–153.

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